

```
while count < 3 and guess.lower() != name:    # .lower allows things like
    Guilherme to still match
    print("You are wrong!")
    guess = input("What is my name? ")
    count = count + 1

if guess.lower() != name:
    print("You are wrong!") # this message isn't printed in the third chance, so
    we print it now
    print("You ran out of chances.")
else:
    print("Yes! My name is", name + "!")
```



## 13 Dictionaries

This chapter is about dictionaries. Dictionaries have keys and values. The keys are used to find the values. Here is an example of a dictionary in use:

```
def print_menu():
    print('1. Print Phone Numbers')
    print('2. Add a Phone Number')
    print('3. Remove a Phone Number')
    print('4. Lookup a Phone Number')
    print('5. Quit')
    print()

numbers = {}
menu_choice = 0
print_menu()
while menu_choice != 5:
    menu_choice = int(input("Type in a number (1-5): "))
    if menu_choice == 1:
        print("Telephone Numbers:")
        for x in numbers.keys():
            print("Name: ", x, "\tNumber:", numbers[x])
        print()
    elif menu_choice == 2:
        print("Add Name and Number")
        name = input("Name: ")
        phone = input("Number: ")
        numbers[name] = phone
    elif menu_choice == 3:
        print("Remove Name and Number")
        name = input("Name: ")
        if name in numbers:
            del numbers[name]
        else:
            print(name, "was not found")
    elif menu_choice == 4:
        print("Lookup Number")
        name = input("Name: ")
        if name in numbers:
            print("The number is", numbers[name])
        else:
            print(name, "was not found")
    elif menu_choice != 5:
        print_menu()
```

And here is my output:

```
1. Print Phone Numbers
2. Add a Phone Number
3. Remove a Phone Number
4. Lookup a Phone Number
5. Quit
```

```
Type in a number (1-5): 2
Add Name and Number
```

```
Name: Joe
Number: 545-4464
Type in a number (1-5): 2
Add Name and Number
Name: Jill
Number: 979-4654
Type in a number (1-5): 2
Add Name and Number
Name: Fred
Number: 132-9874
Type in a number (1-5): 1
Telephone Numbers:
Name: Jill    Number: 979-4654
Name: Joe    Number: 545-4464
Name: Fred    Number: 132-9874

Type in a number (1-5): 4
Lookup Number
Name: Joe
The number is 545-4464
Type in a number (1-5): 3
Remove Name and Number
Name: Fred
Type in a number (1-5): 1
Telephone Numbers:
Name: Jill    Number: 979-4654
Name: Joe    Number: 545-4464

Type in a number (1-5): 5
```

This program is similar to the name list earlier in the chapter on lists. Here's how the program works. First the function `print_menu` is defined. `print_menu` just prints a menu that is later used twice in the program. Next comes the funny looking line `numbers = {}`. All that this line does is to tell Python that `numbers` is a dictionary. The next few lines just make the menu work. The lines

```
for x in numbers.keys():
    print("Name:", x, "\tNumber:", numbers[x])
```

go through the dictionary and print all the information. The function `numbers.keys()` returns a list that is then used by the `for` loop. The list returned by `keys()` is not in any particular order so if you want it in alphabetic order it must be sorted. Similar to lists the statement `numbers[x]` is used to access a specific member of the dictionary. Of course in this case `x` is a string. Next the line `numbers[name] = phone` adds a name and phone number to the dictionary. If `name` had already been in the dictionary `phone` would replace whatever was there before. Next the lines

```
if name in numbers:
    del numbers[name]
```

see if a name is in the dictionary and remove it if it is. The operator `name in numbers` returns true if `name` is in `numbers` but otherwise returns false. The line `del numbers[name]` removes the key `name` and the value associated with that key. The lines

```
if name in numbers:
    print("The number is", numbers[name])
```

check to see if the dictionary has a certain key and if it does prints out the number associated with it. Lastly if the menu choice is invalid it reprints the menu for your viewing pleasure.

A recap: Dictionaries have keys and values. Keys can be strings or numbers. Keys point to values. **Values** can be any type of variable (including lists or even dictionaries (those dictionaries or lists of course can contain dictionaries or lists themselves (scary right? :-))) ). Here is an example of using a list in a dictionary:

```
max_points = [25, 25, 50, 25, 100]
assignments = ['hw ch 1', 'hw ch 2', 'quiz', 'hw ch 3', 'test']
students = {'#Max': max_points}

def print_menu():
    print("1. Add student")
    print("2. Remove student")
    print("3. Print grades")
    print("4. Record grade")
    print("5. Print Menu")
    print("6. Exit")

def print_all_grades():
    print('\t', end=' ')
    for i in range(len(assignments)):
        print(assignments[i], '\t', end=' ')
    print()
    keys = list(students.keys())
    keys.sort()
    for x in keys:
        print(x, '\t', end=' ')
        grades = students[x]
        print_grades(grades)

def print_grades(grades):
    for i in range(len(grades)):
        print(grades[i], '\t', end=' ')
    print()

print_menu()
menu_choice = 0
while menu_choice != 6:
    print()
    menu_choice = int(input("Menu Choice (1-6): "))
    if menu_choice == 1:
        name = input("Student to add: ")
        students[name] = [0] * len(max_points)
    elif menu_choice == 2:
        name = input("Student to remove: ")
        if name in students:
            del students[name]
        else:
            print("Student:", name, "not found")
    elif menu_choice == 3:
        print_all_grades()
    elif menu_choice == 4:
        print("Record Grade")
        name = input("Student: ")
        if name in students:
            grades = students[name]
            print("Type in the number of the grade to record")
            print("Type a 0 (zero) to exit")
            for i in range(len(assignments)):
                print(i + 1, assignments[i], '\t', end=' ')
            print()
            print_grades(grades)
            which = 1234
            while which != -1:
                which = int(input("Change which Grade: "))
                which -= 1 #same as which = which - 1
                if 0 <= which < len(grades):
```

```

        grade = int(input("Grade: "))
        grades[which] = grade
    elif which != -1:
        print("Invalid Grade Number")
    else:
        print("Student not found")
elif menu_choice != 6:
    print_menu()

```

and here is a sample output:

```

1. Add student
2. Remove student
3. Print grades
4. Record grade
5. Print Menu
6. Exit

Menu Choice (1-6): 3
      hw ch 1      hw ch 2      quiz      hw ch 3      test
#Max   25         25         50         25         100

Menu Choice (1-6): 5
1. Add student
2. Remove student
3. Print grades
4. Record grade
5. Print Menu
6. Exit

Menu Choice (1-6): 1
Student to add: Bill

Menu Choice (1-6): 4
Record Grade
Student: Bill
Type in the number of the grade to record
Type a 0 (zero) to exit
1  hw ch 1  2  hw ch 2  3  quiz  4  hw ch 3  5  test
0           0           0           0           0

Change which Grade: 1
Grade: 25
Change which Grade: 2
Grade: 24
Change which Grade: 3
Grade: 45
Change which Grade: 4
Grade: 23
Change which Grade: 5
Grade: 95
Change which Grade: 0

Menu Choice (1-6): 3
      hw ch 1      hw ch 2      quiz      hw ch 3      test
#Max   25         25         50         25         100
Bill   25         24         45         23         95

Menu Choice (1-6): 6

```

Here's how the program works. Basically the variable `students` is a dictionary with the keys being the name of the students and the values being their grades. The first two lines just create two lists. The next line `students = {'#Max': max_points}` creates a new dictionary with the key `{#Max}` and the value is set to be `[25, 25, 50, 25, 100]`